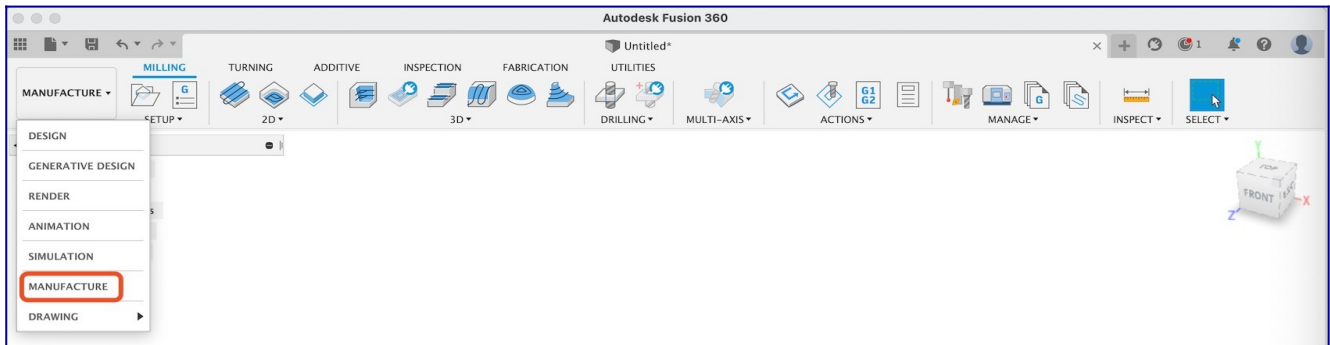


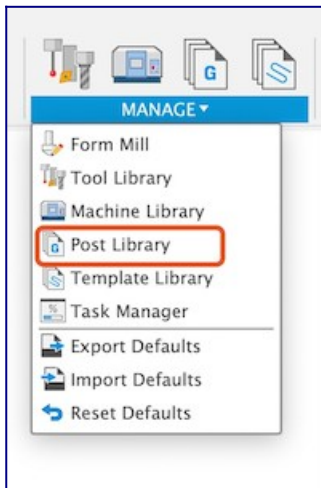
Fusion 360

Install PostProcessor

1. Start Fusion 360.
2. Navigate to the Manufacture Workspace.

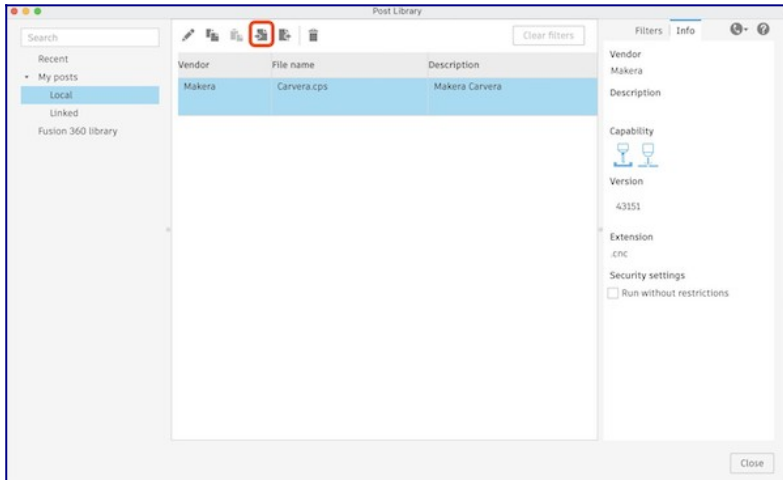


3. Under the Manage group, select the Post Library button.



4. Navigate to the "My Posts->Local" folder.

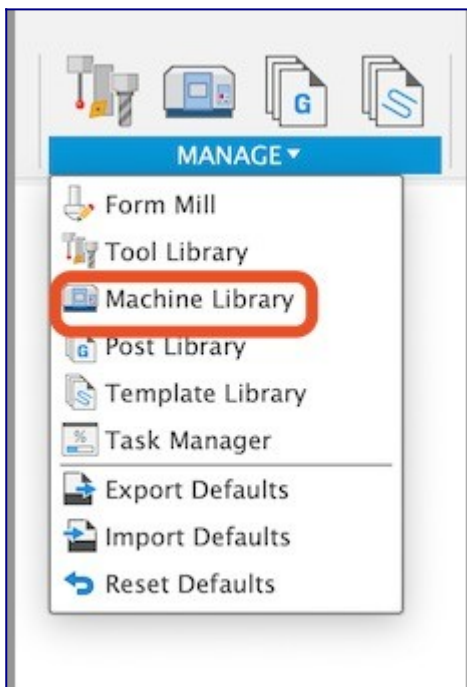
5. Click the "Import" button.



6. Select the "Carvera.cps" file and finish the importing.

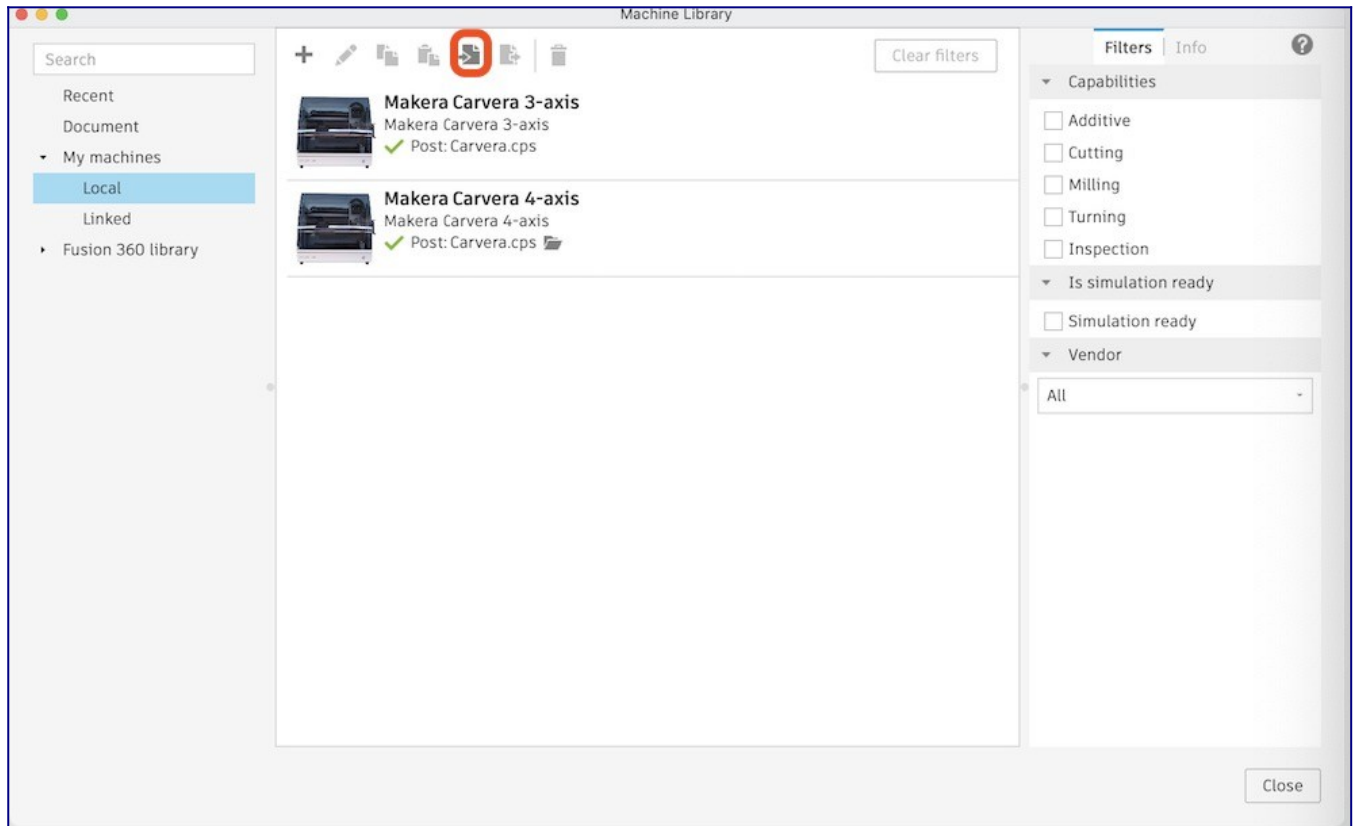
Install Machine

1. Under the Manage group, select the Machine Library button.



2. Navigate to the "My Machines->Local" folder.

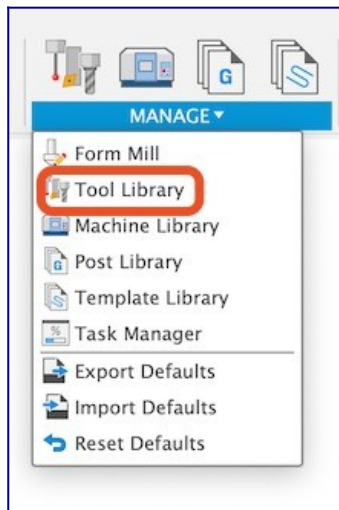
3. Click the "Import" button.



4. Import the machine definition .f3d files from CAM_Post_Processors/Fusion360-profiles/. They are labeled by machine and features

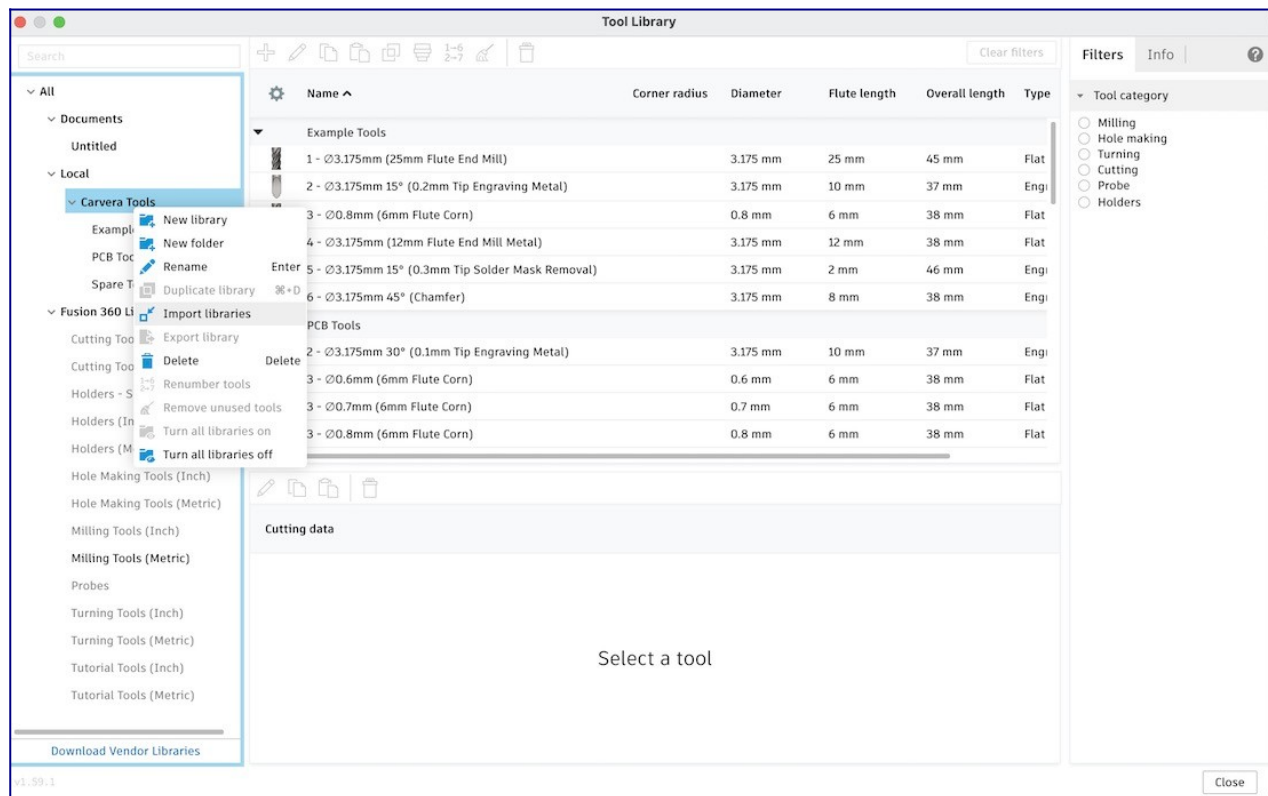
Install Tool Library

1. Under the Manage group, select the Tool Library button.



2. Navigate to "All->Local".

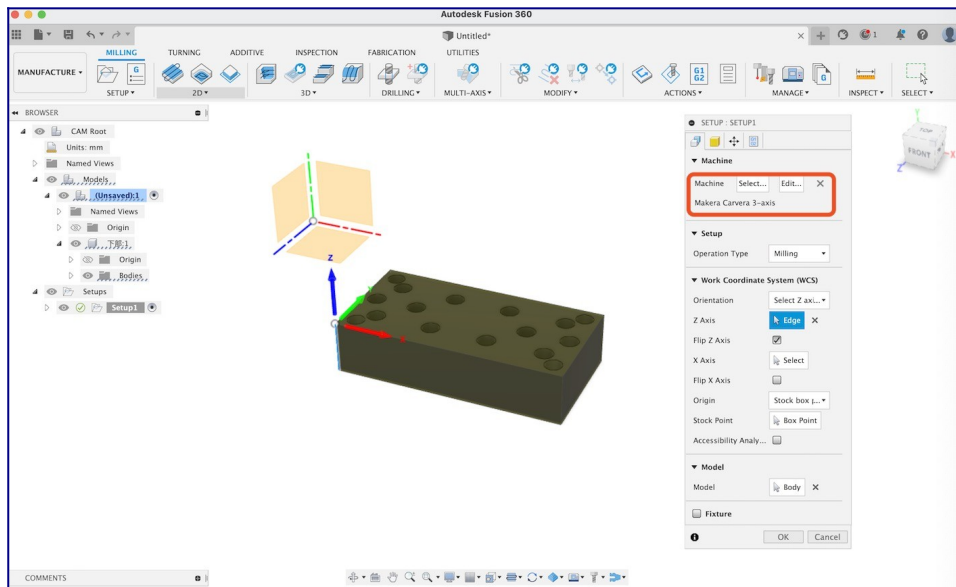
- Right click the "Local" folder and click the "New folder" button, create a new folder named "Carvera Tools".
- Right click the new folder and then click the "Import libraries" button.



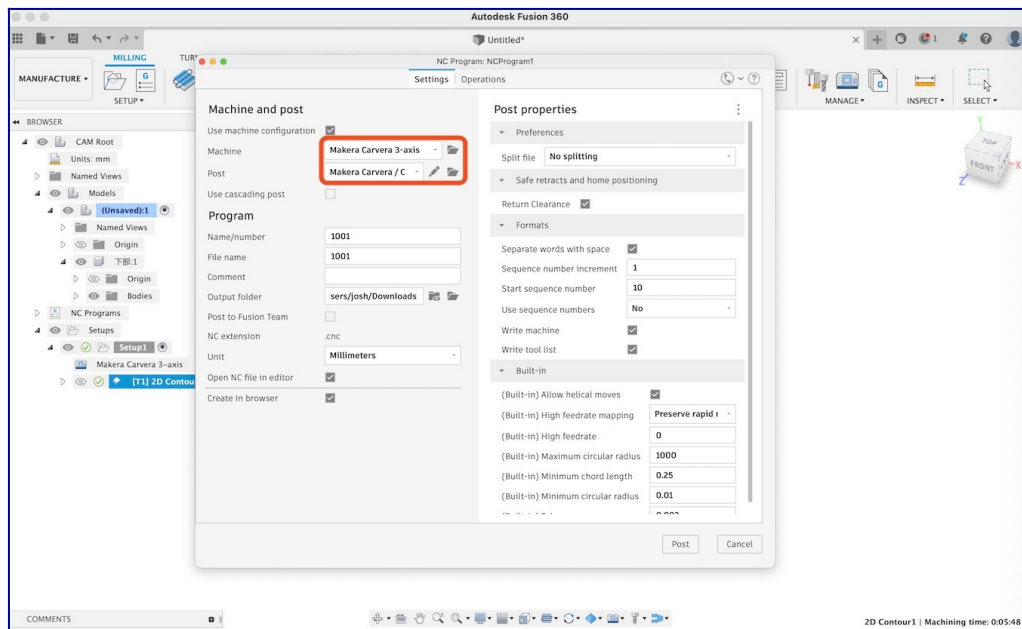
- Import the .tools files from CAM_Post_Processors/Fusion360-profiles/Tool Files/ (for example: Makera Ball Endmills.tools, Makera Drill Bits.tools, Makera Engraving Bits.tools, Makera 0 Flute Bits.tools, Makera Corn Bits.tools, Makera Thread Mills.tools, Makera Other Tools.tools, and Makera ToolHolders.tools).

Using Fusion360 Profiles

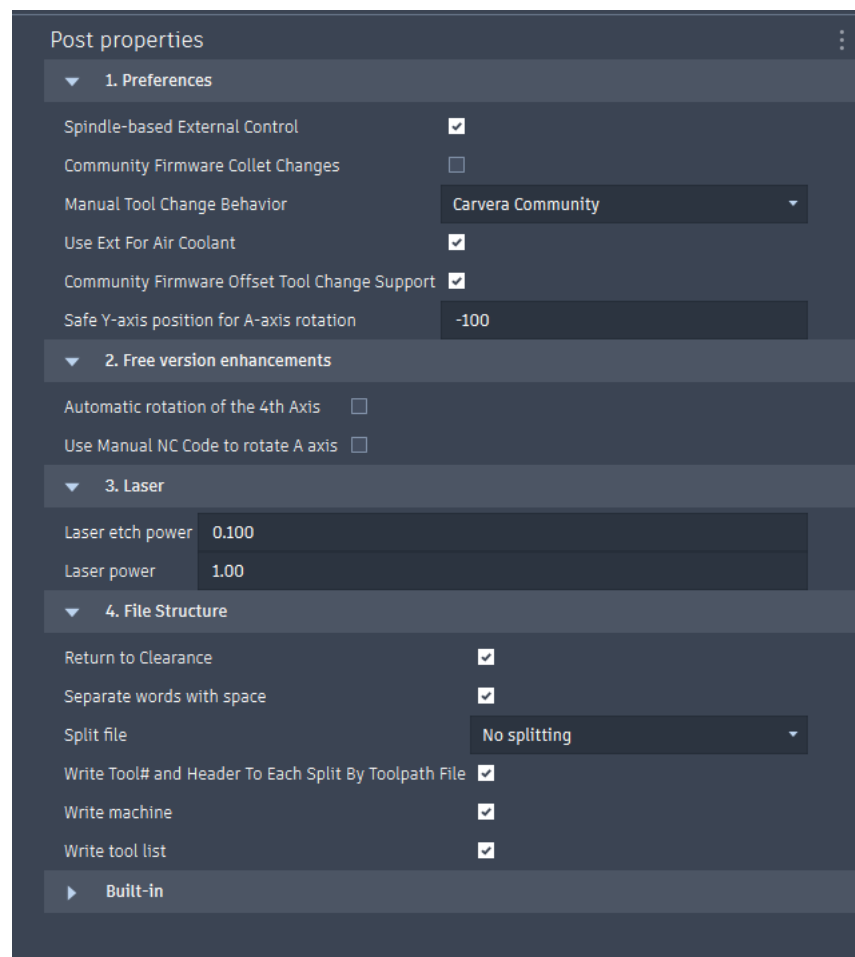
1. When creating setups, select the imported Makera Carvera machine profile that matches your hardware (for example 3-axis, 4-axis, Air, or Harmonic).



2. When doing post process, select the Makera Carvera machine and the Carvera.cps post file.



3. the post properties page now looks like this:



The screenshot shows a 'Post properties' window with a dark theme. It is organized into sections with expandable headers. The first section, '1. Preferences', contains several settings: 'Spindle-based External Control' (checked), 'Community Firmware Collet Changes' (unchecked), 'Manual Tool Change Behavior' (set to 'Carvera Community'), 'Use Ext For Air Coolant' (checked), 'Community Firmware Offset Tool Change Support' (checked), and 'Safe Y-axis position for A-axis rotation' (set to '-100'). The second section, '2. Free version enhancements', has 'Automatic rotation of the 4th Axis' (unchecked) and 'Use Manual NC Code to rotate A axis' (unchecked). The third section, '3. Laser', has 'Laser etch power' (0.100) and 'Laser power' (1.00). The fourth section, '4. File Structure', has 'Return to Clearance' (checked), 'Separate words with space' (checked), 'Split file' (set to 'No splitting'), 'Write Tool# and Header To Each Split By Toolpath File' (checked), 'Write machine' (checked), and 'Write tool list' (checked). At the bottom, there is a 'Built-in' section with a right-pointing arrow.

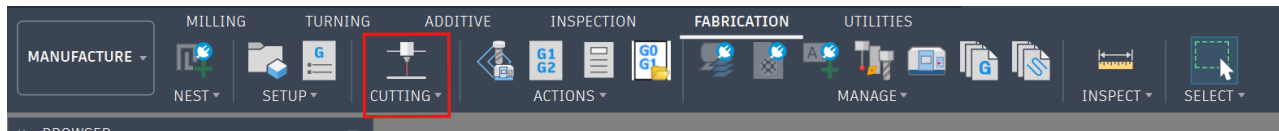
Section	Property	Value
1. Preferences	Spindle-based External Control	☒
	Community Firmware Collet Changes	☐
	Manual Tool Change Behavior	Carvera Community
	Use Ext For Air Coolant	☒
	Community Firmware Offset Tool Change Support	☒
	Safe Y-axis position for A-axis rotation	-100
2. Free version enhancements	Automatic rotation of the 4th Axis	☐
	Use Manual NC Code to rotate A axis	☐
3. Laser	Laser etch power	0.100
	Laser power	1.00
4. File Structure	Return to Clearance	☒
	Separate words with space	☒
	Split file	No splitting
	Write Tool# and Header To Each Split By Toolpath File	☒
	Write machine	☒
	Write tool list	☒
Built-in		

You want to select the machine you are using in the manual tool change behavior.
Spindle based external control turns on the ext port whenever the spindle is turned on
Ext for air coolant turns on the ext port whenever the coolant type is set to mist
All of the community options are expanded upon in the community gitbook: [Gitbook - View](#)

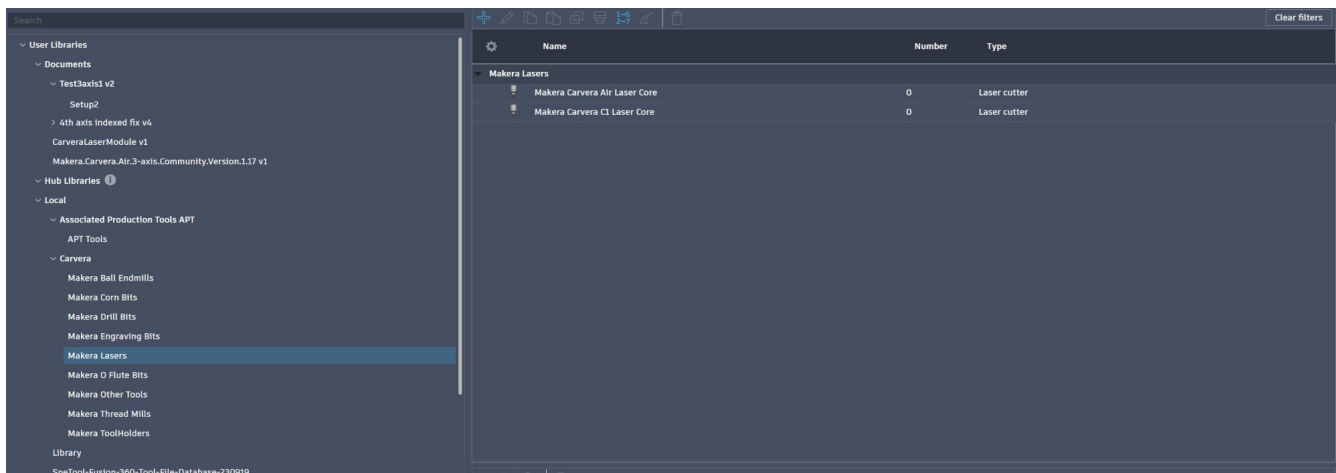
The laser module only works with one set of etch and engraving values. It can be set up in the following way:

Using the laser:

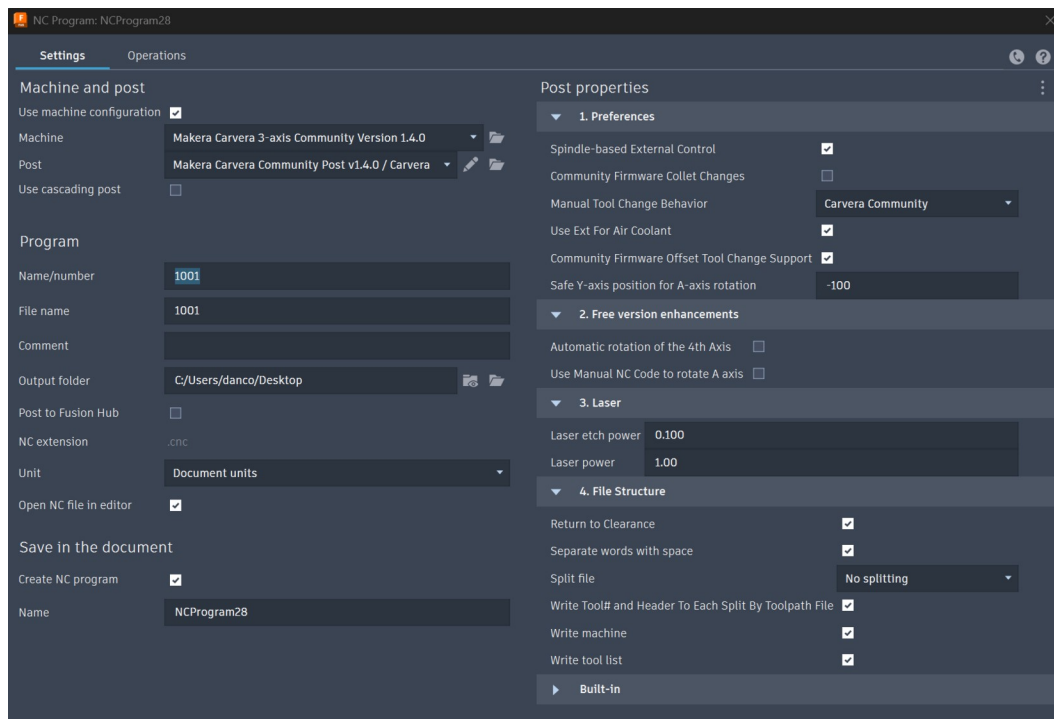
1. in the fabrication manufacturing tab, select the laser cutting feature



2. Select the carvera or carvera air laser from the tools pack attached to this github



3. use through-auto or etch cutting method
4. when posting the file set the power you want the laser to use, laser power is for through cutting



Manual NC dropdown commands:

Dwell -pause for x seconds

Stop -pause and wait for input from the user

Comment -write a comment into the file

Measure Tool -run a tool length offset calibration on the current tool

Tool Break Control -run a tool break test. Requires the community firmware

Pass Through -send the contents of the input box directly to the machine

Useful Pass Through Commands

M98.1 "nameOfFile" -open subprogram by name

M98 P2002 -open subprogram by number, this command would open and run /macros/2002.cnc before returning to the main program

The following ACTION commands are supported by this post.

RapidA:# -rapids the a axis to degree position

SaferA:# -Moves the z axis up to its clearance position then moves the a axis

SafeZ -Go to a safe z height (same height as the clearance position)

SpindleOff -turns the spindle off

Clearance -goes to carvea clearance position

ClearAutoLevel -clears the auto level data from the machine

ResetFeedOverride -resets the spindle override value to 100%

FeedOverride:# -sets the feed override to a given percent. Useful for vetting new programs as well as speeding up an entire set of operations quickly

AirOn -Turns the compressed air on

AirOff -Turns the compressed air off

VacOn -Turns on the vacuum

VacOff -Turns off the vacuum

AutoVacOn -turns on auto vacuum

AutoVacOff -turns off auto vacuum

LightOn -turns on the light

LightOff -turns off the light

ShrinkA -Shrinks the A axis with offset 0, so A365 will turn into A5